

MATHEMATICS XI

10 Year Questions Distribution

According to the Chapters.

This Distribution is carried out According to
NEW BOOK OF MATHEMATICS.

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QUESTIONS		Ex. No.	YEARS
Unit: 1	Complex Numbers		
Section B			
2. i) Separate the real and imaginary parts of $(\sqrt{2} + i)^{-2}$.		1.2	2025 P.E.
2. (i) Find the real and imaginary parts of $(1 - \sqrt{5}i)^{-1}$.		1.2	2025 S. G.
2. i) Solve the following complex equation. $(x, y)(2, 3) = (-4, 7)$		1.1	2024 P.E.
2. i) Find the real and imaginary parts of the $\left(\frac{1-i}{2-i}\right)^{-1}$		1.2	2024 S. G.
EXAMPLE (II), TOPIC 1.2.5 type (II)			
2. i) Solve the quadratic equation: $z^2 - 4z + 5 = 0$ by completing the squares where z is a complex number.		1.3	2023
Section C			
Unit: 2	Matrices and Determinants		
Section B			
ii) Using properties, Prove that:		2.4	2025 P.E.
$\begin{vmatrix} \alpha & \beta\gamma & \alpha\beta\gamma \\ \beta & \gamma\alpha & \alpha\beta\gamma \\ \gamma & \alpha\beta & \alpha\beta\gamma \end{vmatrix} = \begin{vmatrix} \alpha & \alpha^2 & \alpha^3 \\ \beta & \beta^2 & \beta^3 \\ \gamma & \gamma^2 & \gamma^3 \end{vmatrix}$			

iii) Show that $\begin{bmatrix} 1 & -2 & -6 \\ -3 & 2 & 9 \\ 2 & 0 & -3 \end{bmatrix}$ is a periodic matrix of period 2.	2.1	
ii) Using the properties of determinant show that: $\begin{vmatrix} 1 & x & y+z \\ 1 & y & z+x \\ 1 & z & x+y \end{vmatrix} = 0.$	2.4	2025 S.G.
iii) Find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 2 & 2 \end{bmatrix}$ using elementary row operations.	2.5	
ii) If $A = \begin{bmatrix} -2 & 1 & 0 \\ -1 & 4 & 3 \\ 0 & 8 & 5 \end{bmatrix}$ then find $A^2 - 5A + 4I$	2.2	2024 P.E.
iii) Using properties of determine prove that: $\begin{vmatrix} b+c & a & a \\ b & c+a & b \\ c & c & a+b \end{vmatrix} = 4abc$	2.4	
ii) Show that the $A = \begin{bmatrix} -5 & -8 & 0 \\ 3 & 5 & 0 \\ 1 & 2 & -1 \end{bmatrix}$ is an involutory matrix.:	2.1	2024 S.G.
EXAMPLE:, TOPIC 2.1.3 part (vii)		
iii) Without expanding show that $\begin{vmatrix} x+1 & x+3 & x+5 \\ x+4 & x+6 & x+8 \\ x+7 & x+9 & x+11 \end{vmatrix} = 0$	2.4	
ii) Find the real numbers x, y, z such that matrix A is Hermitian matrix, where $A = \begin{bmatrix} 3 & x+2i & yi \\ 3-2i & 0 & 1+zi \\ yi & i-xi & -1 \end{bmatrix}$	2.1	2023
iii) Without expanding determinants, prove that $\begin{vmatrix} \alpha & \beta\gamma & \alpha\beta\gamma \\ \beta & \gamma\alpha & \alpha\beta\gamma \\ \gamma & \alpha\beta & \alpha\beta\gamma \end{vmatrix} = \begin{vmatrix} \alpha & \alpha^2 & \alpha^3 \\ \beta & \beta^2 & \beta^3 \\ \gamma & \gamma^2 & \gamma^3 \end{vmatrix}$	2.4	
Section C		
3. Use Gauss-Jordan method to solve the system of linear equation:	2.6	2025 P.E.

$\begin{aligned} x - y + 4z &= 4 \\ 2x + 2y - z &= 2 \\ 3x - 2y + 3z &= -3 \end{aligned}$		
3. Use Gauss-Jordan method to solve the system of linear equations: $x + 5y + 2z = 9, x + y + 7z = 6, -3y + 4z = -2$ EXAMPLE:, TOPIC 2.6.4 part (iv)	2.6	2025 S. G.
3. Solve the non-homogenous system of linear equations using Gauss Elimination method: $-x + y + z = 0, x + 2y = \dots, -3x + 2y - z = -2$	2.6	2024 P.E.
3. Use Gauss, Jordan method to solve the system of linear equations: $2x + 2y - z = 4, x - 2y + z = 2, x + y = 0$	2.6	2024 S. G.
3. Solve the following non-homogenous system of linear equations using Gauss Jordan – Method. $\begin{aligned} x - y + 4z &= 4 \\ 2x + 2y - z &= 2 \\ 3x - 2y + 3z &= -3 \end{aligned}$	2.6	2023

Unit: 3

Vectors

Section B

iv) Find the angle between the vectors $3\hat{i} + 5\hat{j} - 3\hat{k}$ and $\hat{i} - 4\hat{j} + \hat{k}$.	3.4	2025 P.E.
iv) For what value of λ if the vector $\hat{i} + \hat{j} + 2\hat{k}, \lambda\hat{i} - \hat{j} + \hat{k}$ and $3\hat{i} - 2\hat{j} - \hat{k}$ are coplanar.	3.3	2025 S. G.
iv) Calculate the projection of $\vec{b}$ along $\vec{a}$ when $\vec{a} = \hat{i} - 2\hat{j} + 3\hat{k}$ and $\vec{b} = 3\hat{i} - \hat{j} - 5\hat{k}$	3.4	2024 P.E.
iv) If α, β, γ are the direction angles of a vector then show $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$.	3.4	2024 S. G.
iv) Find λ if the vectors $\hat{i} + \hat{j} + 2\hat{k}, \lambda\hat{i} - \hat{j} + \hat{k}$ and $3\hat{i} - 2\hat{j} - \hat{k}$ are coplanar.	3.3	2023

Section C

4. In a parallelogram $ABCD$, X is the mid point of $\overline{AB}$ and Y divides $\overline{BC}$ in 1:2, Show that if Z divides $\overline{DX}$ in 6:1, then it also divides $\overline{AY}$ in 3:4.	3.1	2025 P.E.
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4. A, B, C are the points of $\vec{a}, \vec{b}$ and $2\vec{a} - \vec{b}$ respectively. D divides $\overline{AC}$ in 2 : 3 and E divides $\overline{BD}$ in 4 : 1. Find the position vector of $\vec{E}$.	3.1	2025 S. G.
7. Find the volume of the tetrahedron whose vertices are $A(2, 1, 8), B(3, 2, 9), C(2, 1, 4)$ and $D(3, 3, 10)$.	3.6	2024 P.E.
7. (OR) Prove Hexagon Law of vector addition.	3.1	
7. The points $A(1, 1), B(5, 4), C(8, 9)$ and $D(0, 3)$ represent the position vectors. Show that the $ABCD$ is a Trapezium.	3.1	2024 S. G.
9. A, B, C are the points $\vec{a}, \vec{b}$ and $2\vec{a} - \vec{b}$ respectively. D divides $\overline{AC}$ in 2 : 3 and E divides $\overline{BD}$ in 4 : 1. Find the position vector of E .	3.1	2023

Unit: 4	Sequence and Series		
Section B			
v) Insert three arithmetic means between 3 and 11.	4.3	2025	P.E.
vi) In a G.P. $a_3 = 12$ and $a_8 = 384$. Find the n^{th} term.	4.5		
EXAMPLE:2 Topic 4.5.2			
v) Find the 13 th term of an A.P. where first term and the common difference are 3 and -4 respectively. Also write its first four terms.	4.2	2025	S. G.
EXAMPLE:3 Topic 4.2.3			
vi) Find the 8 th term of the H.P. $\frac{4}{5}, \frac{2}{5}, \frac{4}{15}, \dots$	4.8		
EXAMPLE:2 Topic 4.8.2			
v) Find Arithmetic, Geometric and Harmonic means between 1 and 9. Also verify that $AH = G^2$	4.9	2024	P.E.
vi) Find three consecutive numbers in G.P whose sum is 39 and their product is 729.	4.5	2024	
EXAMPLE:2 Topic 4.5.2			
v) Find n so that $\frac{x^{n+5} + y^{n+5}}{x^{n+4} + y^{n+4}}$ may be A.M between x and y .	4.3	2023	
Section C			

4. If the 3 rd term and the 7 th term of a H.P are $\frac{2}{9}$ and $\frac{2}{25}$ respectively, find the sequence.	4.8	2024 P.E.
EXAMPLE:3 Topic 4.8.2		
4. If the p^{th} term of a H.P is q , the q^{th} term is p , then the $(p + q)^{\text{th}}$ term is $\frac{pq}{p+q}$.	4.8	2024 S. G.

Unit: 5	Miscellaneous and Series		
Section B			
vii) Find the sum of the series $\sum_{k=2}^{\infty} \left(\frac{1}{k^2-1} \right)$.	5.4	2025	P.E.
vii) Find the sum of series $\sum_{k=1}^{\infty} \frac{1}{4k^2-1}$	5.4	2025	S.G.
vi) Find the sum of the series $3 + 6 + 21 + 96 + 471 + \dots$ to n terms.	5.3	2024	P.E.
vi) Sum the series $1 + 2.5 + 3.5^2 + 4.5^3 + \dots$ to 30 terms.	5.2	2024	S.G.
vi) Find the sum of the series $3 + 6 + 15 + 42 + 123 + \dots$ to n terms.	5.3	2023	
Section C			

Unit: 6	Permutation, Combination And Probability		
Section B			
viii) If ${}^nP_3 = 12$, ${}^n P_3$ Find n .	6.2	2025 P.E.	
EXAMPLE:5, TOPIC 6.2.4			
viii) There are 11 men and 9 women members of a club. How many communities of 8 members can be formed, having exactly five men?	6.3	2025 S. G.	
vii) How many words can be formed by 3 vowels and 4 consonants out of 5 vowels and 7 consonants?	6.3	2024 P.E.	
vii) (OR) How many arrangements can be formed using all the letters of the word 'MATHEMATICS'?	6.2		

vii) How many distinct permutations of letters of the word 'ESSENTIAL' are possible?	6.2	2024 S. G.
vii) In how many ways can two English books, three chemistry books and four physics books be arranged on a shelf so that all the books of same subject are together?	6.2	2023
Section C		
5. Two events A and B such that $P(A) = \frac{1}{4}$, $P(A/B) = \frac{1}{2}$ and $P(B/A) = \frac{2}{3}$ Find $P(A \cap B)$ and $P(B)$.	6.4	2025 P.E.
5. The king, queens and jack of clubs are removed from a deck of 52 playing cards and then shuffled. A card is drawn from remaining cards. Find the probability of getting: (a) a queen (b) a club (c) '9' of red colour	6.4	2025 S. G.
5. In a three child family, find the probability of: (a) Three boys (b) Exactly two boys (c) At least one girl	6.4	2024 P.E.
5. The king, the queen and the jack of clubs are removed from a deck of 52 playing cards and the shuffled. A card is drawn from the remaining cards. Find the probability of getting: (i) A heart (ii) A Queen (iii) A Club	6.4	2024 S. G.
4. A room has 3 lamps from a collection of 10 light bulbs of which 6 are no good, a person selects 3 at random and puts them in the sockets. What is the probability that he will light?	R.E	2023

Unit: 7	Mathematical Induction and Binomial Theorem		
Section B			
ix) Prove that by using principle of mathematical induction $(7^n - 4^n)$ divisible by 3.	7.1	2024 P.E.	
ix) ---OR--- Prove that by using principle of mathematical induction. $1^3 + 2^3 + 3^3 + \dots + n^3 = \left[\frac{n(n+1)}{2} \right]^2$, for all natural numbers n .	7.1		

EXAMPLE:5, TOPIC 7.1.2		
iv) The principle of mathematical induction, prove that $2 + 4 + 6 + \dots + 2n = n(n + 1), \forall n \in \mathbb{N}$	7.1	2024 S. G.
EXAMPLE:1, TOPIC 7.1.2		
viii) Using the principle of mathematical induction for every positive integer n , prove that: $\frac{1}{1.2} + \frac{1}{2.3} + \frac{1}{3.4} + \dots + \frac{1}{n(n+1)} = \frac{n}{n+1}$	7.1	2024 P.E.
ix) Find the middle term in the expansion of $\left(\frac{a}{b} - \frac{b}{a}\right)^{18}$ ix) (OR) Find the term independent of 'x' in $\left(2x + \frac{1}{x^2}\right)^9$	7.2	
viii) Prove the proposition by mathematical induction for every positive integer n . $1 + 4 + 7 + \dots + (3n - 2) = \frac{n(3n - 1)}{2}$	7.1	2024 S. G.
ix) Find the middle term in the expansion of $\left(\frac{a}{h} - \frac{b}{a}\right)^{18}$ ix) (OR) Use binomial theorem to find the value of $(19)^4$.	7.2	
viii) Prove the proposition by mathematical induction for every positive integer $n : 2 + 4 + 6 + \dots + 2n = n(n + 1)$	7.1	2023
EXAMPLE:1, TOPIC 7.1.2		
viii) (OR) Write in the simplified form the term independent of x in the expansion of $\left(2x + \frac{1}{x^2}\right)^9$.	7.2	
Section C		
6. If $x = \frac{1}{3} + \frac{1.3}{3.6} + \frac{1.3.5}{3.6.9} + \dots$ then prove that $x^2 + 2x - 2 = 0$.	7.3	2024 P.E.
EXAMPLE:2, TOPIC 7.3.4		
6. If $x = \frac{1}{3} + \frac{1.3}{3.6} + \frac{1.3.5}{3.6.9} + \dots$ then prove that $x^2 + 2x - 2 = 0$.	7.3	2024 S. G.

EXAMPLE:2, TOPIC 7.3.4		
6. Use Binomial Theorem to show $1 + \frac{1}{4} + \frac{1-3}{4-6} + \frac{1-3-5}{4-6-8} + \dots = \sqrt[3]{4}$	7.3	2024 P.E.
6. Use Binomial Theorem to show that $4x^2 - 2x - 1 = 0$ If $\frac{1}{x} = \frac{2}{5} + \frac{1 \cdot 3}{2!} \left(\frac{2}{5}\right)^2 + \frac{1 \cdot 3 \cdot 5}{3!} \left(\frac{2}{5}\right)^3 + \dots$	7.3	2024 S.G.
5. If $\frac{1}{x} = \frac{2}{5} + \frac{1-3}{2!} \left(\frac{2}{5}\right)^2 + \frac{1-3-5}{3!} \left(\frac{2}{5}\right)^3 + \dots$ then show that $4x^2 - 2x - 1 = 0$.	7.3	2023

Unit: 8	Functions And Graph	
Section B		
x) Find inverse of f and determine domain and range f^{-1} for the real valued function $f(x) = \frac{x-1}{x-3}, x \neq 3$.	8.2	2025 P.E.
x) Find the inverse of the real valued function defined by $f(x) = 5x + 7$ and verify it.	8.2	2025 S.G.
x) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = \sqrt{x+7}; x \geq -7$, find the value of $f^{-1}(x)$	8.2	2024 P.E.
xv) Find the point of intersection of the following functions graphically: $f(x) = x + 4$ and $g(x) = x^2 - 6x + 10$	8.4	
x) Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function defined by $f(x) = -5x + 1$, find the value of $f^{-1}(0)$.	8.2	2024 S.G.
xv) Find the equation of the graph of the function of the type $y = ax^2 + bx + c$ which crosses the x -axis at the points $(-4, 0)$ and $(3, 0)$ and also passes through the point $(2, -4)$.	8.3	
xi) Find the point of intersection of the function $f(x) = x + 4$ and $g(x) = x^2 - 6x + 10$ graphically.	8.4	2023
Section C		
9. Find the point of intersection of the functions $y = 2x + 1$ and $y = x^2 - 4x + 6, \forall x \in \mathbb{R}$ graphically.		2025 P.E.

EXAMPLE:1, TOPIC 8.4.1 (b)		
9. Find the points of intersection of the functions $y = f(x) = 2x + 1$ and $y = g(x) = x^2 - 4x + 6$; $\forall x \in \mathbb{R}$ graphically.		2025 S.G.
EXAMPLE:1, TOPIC 8.4.1 (b)		
6. If $f : \mathbb{R} \rightarrow \mathbb{R}$ is the function defined by $f(x) = 5x +$ 7. Find $f^{-1}(x)$ and verify that $f^{-1}[f(x)] = x$.	8.2	2023
EXAMPLE:1, TOPIC 8.2.1		

Unit: 9	Linear Programming (LP)		
Section B			
x) Solve the following linear programming problem by graphical method when $x \leq 0, y \geq 0$ maximize the objective function $Z(x, y) = 10x + 11y$, subject to constraints $2x + 3y \leq 8; 6x + 3y \leq 10$.	9.3	2023	
Section C			
8. Solve the Linear programming by graphical method when $x \geq 0$ and $y \geq 0$ Maximize $Z(x, y) = 30x + 36y$ subject to: $4x + 2y \leq 12; 6x + 5y \leq 20$	9.3	2024 P.E.	
8. Solve the linear programming problem when $x \geq 0$ and $y \geq 0$. Maximize $Z(x, y) = 30x + 36y$, subject to: $4x + 2y \leq 12; 6x + 5y \leq 20$	9.3	2024 S. G.	

Unit: 10	Trigonometric Identities of Sum and Difference of Angles		
Section B			
xi) Prove that: $\frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta} = \tan 5\theta$.	10.4	2025 P.E.	
xi) Show that: $\frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta} = \tan 5\theta$ (by using trigonometric identities)	10.4	2025 S. G.	
xi) If $A + B + C = 180^\circ$ then prove that $\cos\left(\frac{B+C}{2}\right) = \sin\frac{A}{2}$	10.2	2024 P.E.	

xiv) If $\alpha + \beta + \gamma = 180^\circ$, then show that $\cos\left(\frac{\beta+\gamma}{2}\right) = \sin\frac{\alpha}{2}$.	10.2	2024 S.G.
xii) Express $15\sin\theta + 8\cos\theta$ in the form of $r\sin(\theta + \phi)$ where θ and ϕ are in first quadrant.	10.2	2023
Section C		
8. Prove that: $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$	10.4	2025 P.E.
8. If $\sin\alpha = \frac{12}{13}$ and $\sin\beta = \frac{3}{5}$ where α and β lie in the first quadrant then find the values of $\cos(\alpha + \beta)$ and $\sin(\alpha + \beta)$	10.1	2024 S.G.
9. Prove fundamental law $\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$	10.1	2024 P.E.
TOPIC 10.1.2		
9 (OR) Prove that:	10.3	
(i) $\cos 4\theta = 8\cos^4\theta - 8\cos^2\theta + 1$		
(ii) $\frac{\sin 6\theta + \sin 4\theta}{\cos 6\theta + \cos 4\theta} = \tan 5\theta$	10.4	
9. If $\cos\theta = \frac{3}{5}$, where $0 < \theta < \frac{\pi}{2}$, then find the value of $\tan 3\theta$.	10.3	2024 S.G.
7. Prove that $\cos 20^\circ \cos 40^\circ \cos 60^\circ \cos 80^\circ = \frac{1}{16}$ by using trigonometric formulae:	10.4	2023

Unit: 11	Application of Trigonometry		
Section B			
xii) Find the smallest angle in $\triangle ABC$ when $a = 6\text{cm}$, $b = 8\text{cm}$ and $c = 9.4\text{cm}$.	11.2	2025 P.E.	
xiii) Find the value of R and r in $\triangle ABC$ when $a = 40\text{cm}$, $b = 12\text{cm}$ and $c = 38\text{cm}$.	11.4		
xii) The measure of two sides of a triangle are 4 and 5 units. Find the third side so that area of triangle is 6 square units.	11.3	2024 S.G.	
xiii) Prove that: $\frac{1}{r^2} = \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2+b^2+c^2}{\Delta^2}$.	11.4		

xii) Find the smallest angle in $\triangle ABC$ when $a = 25\text{cm}$, $b = 18\text{cm}$ and $c = 21\text{cm}$	11.2	2024 P.E.
xiii) Prove that $\frac{1}{r^2} + \frac{1}{r_1^2} + \frac{1}{r_2^2} + \frac{1}{r_3^2} = \frac{a^2+b^2+c^2}{\Delta^2}$	11.4	
xi) The sides of a parallelogram are 25cm and 35cm long and one of its angle is 36° . Find the length of the diagonal(s).	11.2	2024 S.G.
xii) Prove that: $r = 4R \sin \frac{\alpha}{2} \sin \frac{\beta}{2} \sin \frac{\gamma}{2}$	11.4	
ix) Find the measure of the largest angle in $\triangle ABC$ with $a = 10\text{cm}$, $b = 20\text{cm}$ and $c = 26\text{cm}$. EXAMPLE:3, TOPIC 11.1.3	11.2	2023
xiii) Prove that $r_1 r_2 + r_2 r_3 + r_3 r_1 = s^2$.	R.E	

Unit: 12

Graphs of Trigonometric and Inverse Trigonometric Functions and Solution of Trigonometric Equations

Section B

xiv) Find the maximum and minimum values of the function: $y = \frac{1}{4-5 \sin(7\theta-8)}$ EXAMPLE:(iii), TOPIC 12.1.3(b)	12.1	2024 S.G.
xiv) Solve the equation $\cos \theta - \theta = 0$ graphically for interval $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. EXAMPLE:2, TOPIC 12.3.2	12.3	2024 P.E.
xiv) OR Show that $2 \tan^{-1} \left(\frac{2}{3} \right) = \cos^{-1} \left(\frac{5}{13} \right)$	12.4	
xiii) Draw the graph of $y = -\sin \theta$; where $0 \leq \theta \leq \pi$	12.2	2024 S.G.
xiii) (OR) Show that $\cos^{-1}(y) + \cos^{-1}(-y) = \pi$	12.4	
xi) (OR) By using graph solve the equation $\cos \theta - \theta = 0$ for the interval $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$. EXAMPLE:2, TOPIC 12.3.2	12.3	2023

xiv) Show that: $\tan^{-1}\left(\frac{3}{4}\right) + \tan^{-1}\left(\frac{3}{5}\right) = \tan^{-1}\left(\frac{8}{19}\right) = \frac{\pi}{4}$	12.4	
Section C		
7. Show that $2 \tan^{-1}\left(\frac{2}{3}\right) = \cos^{-1}\left(\frac{5}{13}\right)$	12.4	2025 P.E.
7. Using formulae of trigonometric function, prove that: $\sin^{-1}\left(\frac{77}{85}\right) + \sin^{-1}\left(\frac{3}{5}\right) = \cos^{-1}\left(\frac{87}{425}\right)$	12.4	2025 S.G.
10. Find the general solution set of the trigonometric equation $\operatorname{cosec} \theta = \sqrt{3} + \cot \theta$ and also verify the solution.	12.5	2024 P.E.
10. Find the general solution of the trigonometric equation $4\cos^2 x - 8\sin x + 1 = 0$. Also verify the solution.	12.5	2024 S.G.
8. Find the general solution of the trigonometric equation $\sin 4\theta - \sin 2\theta - \cos 3\theta = 0$ and verify the solution.	12.5	2023